

Operational Amplifier (Op-Amp)

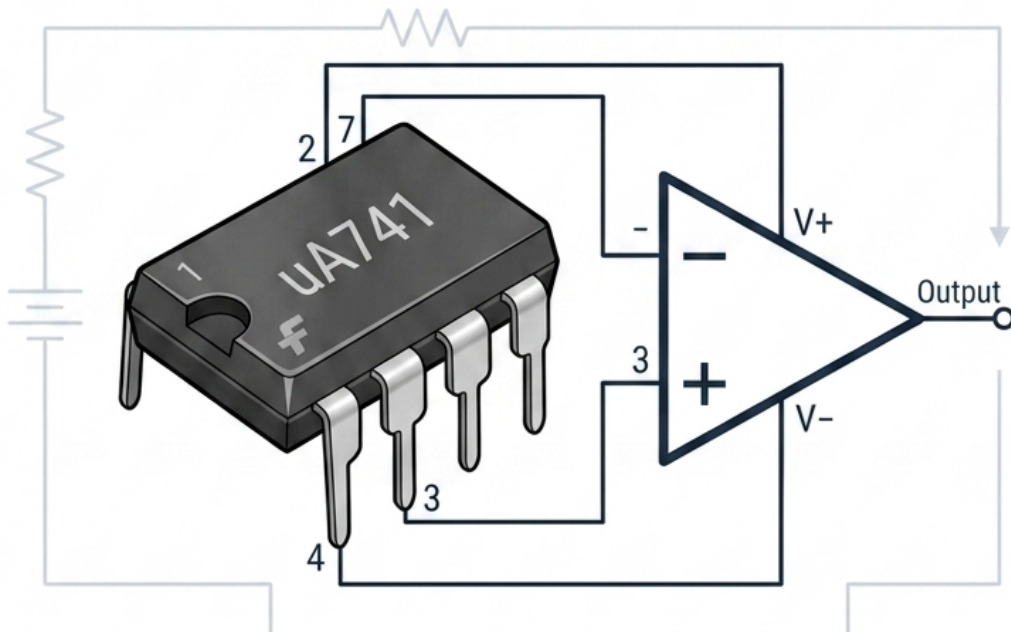
Fundamental Building Block in Analog Electronics

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1. Introduction

An **Operational Amplifier (Op-Amp)** is a high-gain, direct-coupled differential amplifier that forms the cornerstone of modern analog circuit design. Packaged as an integrated circuit, it accepts two voltage inputs and outputs an amplified version of their difference. The term *operational* originates from the device's original use in performing mathematical operations — addition, subtraction, integration, and differentiation — in analog computers of the 1940s and 50s.

Today, Op-Amps are ubiquitous: they appear in audio equipment, medical instrumentation, industrial sensors, communication systems, and power electronics. A single inexpensive IC such as the LM741 or TL071 can replace dozens of discrete transistors, while offering predictable, stable performance across a wide range of applications.

- Extremely high voltage gain ($10^5 - 10^8$ open-loop)
- Very high input impedance (minimizes circuit loading)
- Versatile: the same chip serves in amplifiers, filters, oscillators, and comparators
- Low cost and compact integrated circuit form factor

2. Basic Concept & Differential Amplification

The Op-Amp contains two distinct input terminals and one output terminal. The **inverting input (-)** and **non-inverting input (+)** together define the *differential input voltage*. The device amplifies the *difference* between these two potentials, rejecting any signal common to both (common-mode rejection).

Open-Loop Gain (A_{OL})

When no external feedback network is connected, the Op-Amp operates in *open-loop* mode. The open-loop voltage gain A_{OL} is defined as:

$$V_{out} = A_{OL} \times (V_+ - V_-)$$

Where $A_{OL} \approx 10^5$ to 10^8 for typical Op-Amps (e.g. LM741: $\sim 200,000$)

This enormous gain makes open-loop operation impractical for linear amplification — any micro-volt imbalance drives the output to the supply rail. Instead, **negative feedback** is applied to control and stabilize the gain.

★ Design Note

Always verify the Op-Amp's supply voltage range before powering a circuit. Exceeding the absolute maximum rating permanently damages the device.

3. Internal Structure Overview

Internally, a monolithic Op-Amp IC consists of three cascaded functional stages, all fabricated on a single silicon die:

Stage	Components	Primary Function
① Differential Amplifier	Matched transistor pair (BJT or MOSFET)	Amplifies $V_+ - V_-$; rejects common-mode signals

② Voltage Gain Stage	CE/CS amplifier + compensation capacitor	Provides the bulk of open-loop voltage gain
③ Output Stage	Class AB push-pull emitter follower	Low output impedance; drives external loads

The internal frequency-compensation capacitor (typically ~30 pF in the LM741) limits the gain-bandwidth product, ensuring stability. Modern Op-Amps such as the TL071 or OPA2134 replace BJTs with JFETs for ultra-high input impedance.

4. Circuit Symbol & Pin Description

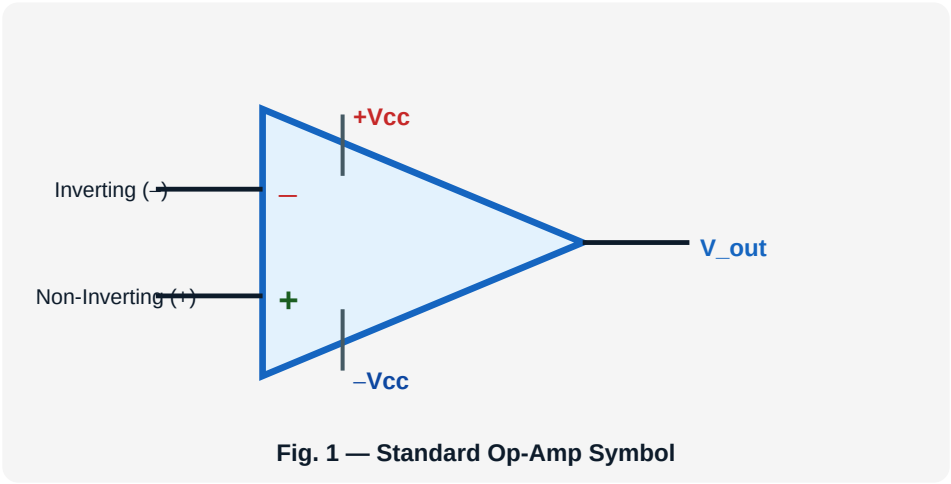


Fig. 1 — Standard Op-Amp Symbol

Pin	Symbol	Description
Inv. Input	V- (-)	Signal applied here is amplified and inverted at output
Non-Inv. Input	V+ (+)	Signal applied here is amplified without phase inversion
Output	V_out	Amplified output voltage: $A \times (V+ - V-)$
Pos. Supply	+Vcc	Positive power supply (e.g. +12 V or +15 V)
Neg. Supply	-Vcc	Negative power supply (e.g. -12 V or -15 V)

5. Ideal Op-Amp Characteristics

For circuit analysis, engineers model the Op-Amp as an *ideal* device with the following properties. Real devices approach but never fully achieve these ideals.

Parameter	Ideal Value	Typical LM741	Effect
Open-Loop Gain A_{OL}	∞	200,000	Precise gain set by feedback only
Input Impedance Z_{in}	∞	2 M Ω	No loading of source circuit
Output Impedance Z_{out}	0 Ω	75 Ω	Drives any load without drop
Bandwidth	∞	1 MHz (GBP)	Flat gain at all frequencies
Offset Voltage V_{os}	0 V	~2 mV	No DC error at output
CMRR	∞	90 dB	Perfect common-mode rejection

6. Working Principle

The Op-Amp continuously monitors the potential difference between its two input terminals and produces an output proportional to that difference multiplied by the open-loop gain A_{OL} :

$$A_v = V_{out} / (V_+ - V_-)$$

V_{out} = output voltage | V_+ = non-inverting input | V_- = inverting input

In a **closed-loop** system (with negative feedback), a fraction of the output voltage is fed back to the inverting input. This opposes any change at the output, greatly reducing the effective gain but dramatically improving stability, linearity, and bandwidth. The *virtual short* concept states that in an ideal closed-loop Op-Amp, the differential input voltage is driven essentially to zero: $V_+ \approx V_-$.

★ Golden Rules of Op-Amp Analysis

The 'Golden Rules' of ideal Op-Amp analysis: (1) No current flows into either input terminal. (2) The output does whatever is necessary to make $V_+ = V_-$.

7. Common Op-Amp Configurations

7.1 Inverting Amplifier

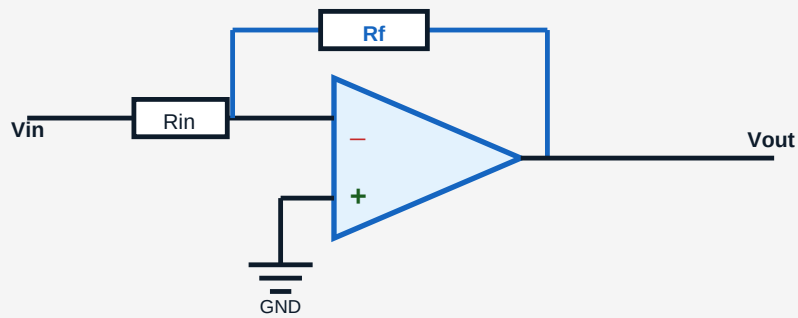


Fig. 2 — Inverting Amplifier Configuration

The input signal is applied to the inverting (–) terminal through resistor R_{in} . The feedback resistor R_f connects the output back to the inverting input. The non-inverting (+) terminal is grounded.

$$V_{out} = -(R_f / R_{in}) \times V_{in}$$

Gain = $-R_f/R_{in}$ | The negative sign indicates 180° phase inversion

- Phase: output is inverted (180°) relative to input
- Closed-loop gain magnitude = R_f / R_{in} (set by resistor ratio)
- Input impedance $\approx R_{in}$ (can be a limitation for high-Z sources)

7.2 Non-Inverting Amplifier

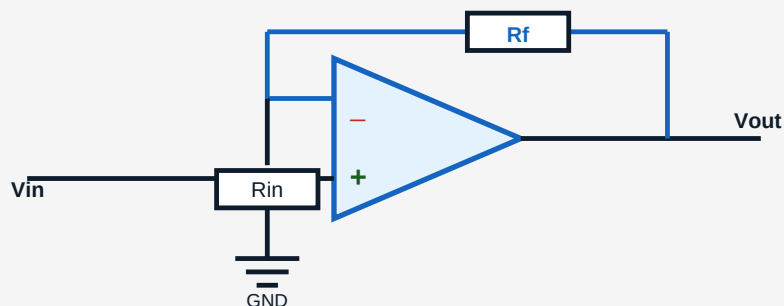


Fig. 3 — Non-Inverting Amplifier Configuration

The input signal connects directly to the non-inverting (+) terminal. The feedback network (R_f and R_{in}) from output to inverting (–) terminal sets the gain.

$$V_{out} = (1 + R_f / R_{in}) \times V_{in}$$

Gain is always ≥ 1 | Output is in-phase with input (0° phase shift)

- Phase: output is in-phase with input — no inversion
- Very high input impedance (ideal for buffering sensors)
- Minimum gain = 1 (when $R_f = 0$ or $R_{in} \rightarrow \infty$)

7.3 Voltage Follower (Unity-Gain Buffer)

A special case of the non-inverting amplifier where $R_f = 0$ and $R_{in} \rightarrow \infty$ (or simply output connected directly to inverting input). The result is unity gain ($A_v = 1$) with maximum input impedance and minimum output impedance.

$$V_{\text{out}} = V_{\text{in}} \quad (\text{Unity Gain})$$

Used for impedance matching — buffers high-Z sources to drive low-Z loads

8. Example Calculations

Example 1 — Inverting Amplifier

Given	Find
$R_{in} = 10\text{ k}\Omega$, $R_f = 47\text{ k}\Omega$, $V_{in} = 0.5\text{ V}$	Output voltage V_{out}

Step 1: Identify the inverting amplifier gain formula:

$$V_{out} = -(R_f / R_{in}) \times V_{in}$$

Step 2: Substitute values:

$$V_{out} = -(47,000 / 10,000) \times 0.5 = -4.7 \times 0.5$$

$$V_{out} = -2.35\text{ V}$$

The output is 2.35 V in amplitude but phase-inverted (negative) relative to input.

Example 2 — Non-Inverting Amplifier

Given	Find
$R_{in} = 5\text{ k}\Omega$, $R_f = 20\text{ k}\Omega$, $V_{in} = 1.2\text{ V}$	Output voltage V_{out} and gain A_v

Step 1: Calculate closed-loop gain:

$$A_v = 1 + (R_f / R_{in}) = 1 + (20\text{k} / 5\text{k}) = 1 + 4 = 5$$

Step 2: Compute output voltage:

$$V_{out} = A_v \times V_{in} = 5 \times 1.2 = 6.0\text{ V}$$

No phase inversion — output is in-phase with the 1.2 V input.

Summary: Configuration Comparison

Configuration	Gain Formula	Phase	Z_{in}	Typical Use
Inverting Amp	$-R_f / R_{in}$	180° inverted	$\approx R_{in}$	Signal processing
Non-Inverting Amp	$1 + R_f / R_{in}$	0° in-phase	Very high	Sensor amplification
Voltage Follower	1 (unity)	0° in-phase	Extremely high	Impedance buffering
Summing Amp	$-(R_f / R_1 + R_f / R_2 + \dots)$	180° inverted	$\approx R_n$	Audio mixing, DAC

Difference Amp	$(R2/R1)(V2-V1)$	Depends	Moderate	Differential sensing
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9. Feedback Concept

Negative feedback is the single most important concept in Op-Amp circuit design. A portion of the output signal is returned to the inverting (–) input, opposing the original input signal. This creates a self-correcting loop with profound benefits:

- **Gain stability:** Closed-loop gain is determined by external resistors (not the chip's internal gain), making it extremely precise and temperature-stable
- **Extended bandwidth:** The gain-bandwidth product (GBP) is constant — reducing gain by negative feedback proportionally increases the usable bandwidth
- **Reduced distortion:** Non-linearities in the open-loop response are divided by the loop gain, greatly improving linearity
- **Controlled impedances:** Feedback can raise input impedance and lower output impedance, optimizing the amplifier for its load

$$\text{Closed-Loop Gain} = A_{OL} / (1 + A_{OL} \times \beta)$$

β = feedback fraction | When $A_{OL} \times \beta \gg 1$, Gain $\approx 1/\beta$ (set by resistors only)

★ Caution: Positive vs. Negative Feedback

Positive feedback is used intentionally in comparators and oscillators to achieve hysteresis and sustained oscillation. However, in amplifiers, any unintended positive feedback will cause instability and oscillation — a common design pitfall.

10. Practical Applications

Application	Configuration	Description
Audio Amplifiers	Non-inverting / Inverting	Amplify microphone or line-level audio signals with low noise
Sensor Conditioning	Instrumentation Amp	Convert tiny sensor outputs (strain gauges, thermocouples) to usable voltage levels
Active Filters	Sallen-Key / MFB	Precisely shape frequency response without inductors
Analog Computation	Summing / Integrator	Perform real-time mathematical operations in control systems
Comparators	Open-loop Op-Amp	Compare two voltages and output a digital HIGH or LOW
Oscillators	Positive feedback	Generate stable sine, square, or triangle waveforms (Wien bridge, astable)
DAC Output Buffer	Voltage Follower	Isolate digital-to-analog converter from load without gain error

11. Advantages & Limitations

✓ Advantages

✗ Limitations

- **High gain:** A_{OL} up to 10^8 enables precise circuits
- **Versatility:** One chip — amplifier, filter, oscillator, comparator
- **Compact:** Entire amplifier stage in an 8-pin DIP IC
- **Low cost:** General-purpose Op-Amps cost under \$0.50
- **Stable with feedback:** Precise gain set by external resistors
- **Low offset:** Precision types offer $V_{os} < 10 \mu V$

- **Bandwidth:** Gain-bandwidth product is fixed (e.g. 1 MHz for LM741)
- **Slew rate:** Output cannot change faster than $\sim 0.5 V/\mu s$ (LM741)
- **Offset voltage:** Non-zero V_{os} causes DC errors
- **Power supply:** Requires dual $\pm V_{cc}$ for rail-to-rail signals
- **Noise:** Internal noise sources limit low-level signal amplification
- **Saturation:** Output clips when driven beyond supply rails

12. Design Tips & Considerations

- **Resistor Values:** Keep R_f and R_{in} in the $1\text{ k}\Omega - 1\text{ M}\Omega$ range. Too low $\rightarrow$ excessive current draw. Too high $\rightarrow$ noise and offset magnification.
- **Power Supply:** Use decoupling capacitors (100 nF ceramic + 10 μ F electrolytic) directly at each supply pin to prevent oscillation from supply noise.
- **Avoiding Saturation:** Ensure input amplitude $\times$ gain $<$ supply voltage. Add output clamping diodes or design with headroom of at least 1–2 V per rail.
- **Frequency Compensation:** For unity-gain-stable Op-Amps (e.g. LM741), no external compensation is needed. For decompensated types (higher speed), add a small capacitor (C_f) across R_f .
- **Noise Reduction:** Choose low-noise Op-Amps (e.g. OPA1612 at 1.1 nV/ $\sqrt{\text{Hz}}$) for audio and sensor work. Shield input traces, minimize PCB trace lengths, and use proper grounding.

13. Real-World Example: Microphone Preamplifier

A dynamic microphone produces a very small signal — typically 1–10 mV. To interface with an audio system requiring a 1 V line level, a preamplifier with a gain of approximately 100 (40 dB) is required.

Design Requirements

- Gain: 100 $\times$ ($A_v = 100$)
- Configuration: Non-inverting amplifier (to preserve phase for stereo imaging)
- Op-Amp: NE5532 (low noise, audio-grade, GBP = 10 MHz)
- Supply: $\pm 15\text{ V}$ dual supply

Component Calculation

Using the non-inverting gain formula:

$$A_v = 1 + R_f / R_{in} \rightarrow 100 = 1 + R_f / R_{in}$$

Choose $R_{in} = 1\text{ k}\Omega$, then:

$$R_f = (100 - 1) \times 1\text{ k}\Omega = 99\text{ k}\Omega \approx 100\text{ k}\Omega \text{ (standard value)}$$

Standard value: $R_{in} = 1\text{ k}\Omega$, $R_f = 100\text{ k}\Omega \rightarrow$ Actual gain = 101

A 10 μ F coupling capacitor at the input blocks DC bias from the microphone, and a 100 Ω output resistor provides short-circuit protection. The NE5532's GBP of 10 MHz ensures flat response well beyond 20 kHz at a gain of 100 (bandwidth $\approx 10\text{ MHz} / 100 = 100\text{ kHz}$ — more than adequate).

★ Verification Check

Always verify that the output swing at full gain does not hit the supply rails. For a 10 mV microphone signal: $10\text{ mV} \times 101 = 1.01\text{ V}$ — well within $\pm 15\text{ V}$.

14. Conclusion

The operational amplifier is one of the most powerful and versatile building blocks in all of electronics engineering. From a single inexpensive integrated circuit, engineers can construct amplifiers, filters, oscillators, comparators, integrators, and signal-conditioning circuits — all by choosing appropriate external components and exploiting the principle of negative feedback.

Key takeaways from this document:

- An Op-Amp amplifies the **difference** between its two input terminals with extremely high open-loop gain
- **Negative feedback** tames the open-loop gain and gives precise, stable, component-determined closed-loop gain
- The **inverting** configuration gives gain $-R_f/R_{in}$ with 180° phase inversion; the **non-inverting** gives $1 + R_f/R_{in}$ with no inversion
- A **voltage follower** (unity-gain buffer) provides near-perfect impedance isolation
- Real Op-Amps have finite bandwidth (GBP), slew rate, and offset voltage — always verify these parameters for your application
- Practical design requires careful attention to supply decoupling, resistor selection, and frequency compensation

Mastering Op-Amp circuit analysis and design is an essential milestone for any electrical or electronics engineering student. The concepts introduced here — differential amplification, feedback theory, and gain-bandwidth trade-offs — underpin virtually every area of modern analog circuit design.